

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	Hard

Question

A sample of oak has a density of **807** kilograms per cubic meter. The sample is in the shape of a cube, where each edge has a length of **0.90** meters. To the nearest whole number, what is the mass, in kilograms, of this sample?

Answer

- A. **588**
- B. **726**
- C. **897**
- D. **1,107**

Correct Answer: A

Rationale

Choice A is correct. It’s given that the sample is in the shape of a cube with edge lengths of **0.9** meters. Therefore, the volume of the sample is **0.90³**, or **0.729**, cubic meters. It’s also given that the sample has a density of **807** kilograms per **1** cubic meter. Therefore, the mass of this sample is **0.729 cubic meters**($\frac{807 \text{ kilograms}}{1 \text{ cubic meter}}$), or **588.303** kilograms. Rounding this mass to the nearest whole number gives **588** kilograms. Therefore, to the nearest whole number, the mass, in kilograms, of this sample is **588**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.